

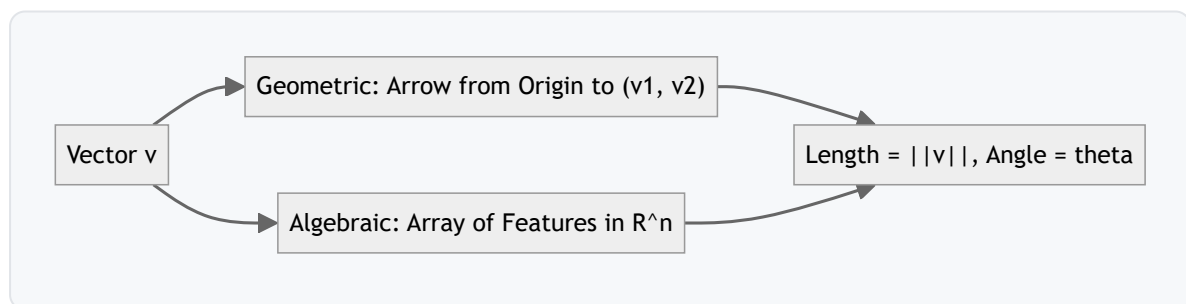
Day 21 - Lecture 21.7: Vectors (Geometric & Algebraic Foundations)

[Python 3.10+](#)[Jupyter Notebook](#)[PDF notes.pdf](#)[Language: English | हिंदी / Hinglish | !\[\]\(e3f8612927870f2e0f9f5989e6dd3064_img.jpg\) Back to Lecture 21 Module](#)

1. Dual Interpretations: Physics vs. Computer Science

A **vector** bridges physical geometry and abstract high-dimensional data representation:

1. **Geometric View (Physics):** A directed line segment characterized by a **magnitude (length)** and a **direction** in space.
2. **Algebraic View (Computer Science / AI):** An ordered tuple of real numbers representing an observation's feature values: $\mathbf{x} = [x_1, x_2, \dots, x_n]^T \in \mathbb{R}^n$.



2. Mathematical Formalism

Column Vector Convention:

In linear algebra, vectors are universally assumed to be vertical column vectors:

$$\mathbf{v} = \begin{bmatrix} v_1 & v_2 & \dots & v_n \end{bmatrix} \in \mathbb{R}^{n \times 1}, \quad \mathbf{v}^T = \begin{bmatrix} v_1 & v_2 & \dots & v_n \end{bmatrix} \in \mathbb{R}^{1 \times n}$$

Vector Norms (Magnitude Measurements):

1. Euclidean Norm (L_2 Norm):

The physical straight-line length from the origin:

$$\|\mathbf{v}\|_2 = \sqrt{\mathbf{v}^T \mathbf{v}} = \sqrt{\sum_{i=1}^n v_i^2}$$

2. Manhattan Norm (L_1 Norm):

Used for Lasso regularization to induce feature sparsity:

$$\|\mathbf{v}\|_1 = \sum_{i=1}^n |v_i|$$

3. Maximum Norm (L_∞ Norm):

$$\|\mathbf{v}\|_\infty = \max_i |v_i|$$

Unit Vectors & Normalization:

A vector with unit length ($|\hat{\mathbf{v}}| = 1.0$) preserving pure direction:

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|_2}$$

3. Python Implementation

```
import numpy as np

# 3D Feature vector
v = np.array([3.0, -4.0, 12.0])

# L2 Norm (Magnitude)
norm_l2 = np.linalg.norm(v, ord=2)

# L1 Norm
norm_l1 = np.linalg.norm(v, ord=1)

# L_inf Norm
norm_linf = np.linalg.norm(v, ord=np.inf)

# Unit Vector
v_unit = v / norm_l2

print(f"Vector v:      {v}")
print(f"L2 Norm (||v||): {norm_l2:.4f} (Exact: sqrt(9 + 16 + 144) = sqrt(169) = 13)")
print(f"L1 Norm:          {norm_l1:.4f} (Exact: 3 + 4 + 12 = 19)")
print(f"L_inf Norm:       {norm_linf:.4f} (Exact: 12)")
print(f"Unit Vector:      {v_unit}")
print(f"Unit Norm:        {np.linalg.norm(v_unit):.4f} (Must be 1.0)")
```

4. Key Takeaways & Interview Points

- **Normalization in NLP:** Word embeddings and transformer vectors are normalized to unit length so that dot product equals cosine similarity ($\mathbf{u} \cdot \mathbf{v} = \cos \theta$).
- **Regularization Connection:** Ridge regression applies an L_2 penalty $|\mathbf{w}|_2^2$; Lasso regression applies an L_1 penalty $|\mathbf{w}|_1$.